

Ontology of Becoming

the argument on one page; the formal core is verified in Lean, the premise is stated

Primitive. A *law* is a map m on a carrier. Write $\text{Fix}m$ for “ m , read as an equation, has a solution” ($\exists x. x = m x$), and $\text{Run}m$ for “ m , read as a recursion, is solved from every seed and every term of every solution differs from its successor.”

Df. m is *structureless* iff m commutes with every relabeling of its carrier’s values. On two values: m commutes with the exchange. On the two-valued carrier this reading coincides with definability, itself a theorem.

L1 (Arena, proved). A carrier with decidable equality admits a structureless law beyond the identity only when it has exactly two values. *[one value admits one map; on three or more, relabeling forces the identity; a moved point and its image exhaust the carrier]*

L2 (Uniqueness, proved). On two values the structureless laws are the identity and negation. The identity holds of everything and singles out nothing; a structureless law that excludes anything is $\neg$.

L3 (Retorsion, proved). A tokening that marks “no distinction is drawn” differently from its denial draws one. A discriminated tokening of the denial makes the denial false.

Th. 1 (No being). $\neg \text{Fix}(\neg)$: each value differs from its own negation, and $X = \neg X$ has no solution.

Th. 2 (Alternation). $\text{Run}(\neg)$; the orbit from each seed is the period-two alternation, unique given the seed, and the two orbits are relabelings of each other.

Th. 3 (Equivalence). $\neg \text{Fix}(\neg) \leftrightarrow \text{Run}(\neg)$. *[the two sides quantify over the same instances; each direction is instantiation]*

Th. 4 (Discrimination). A tokening that marks the denial of occurrence differently from its assertion witnesses occurrence, and the two tokens are distinct. On two values, “one negation apart” and “distinct” coincide, so this is the theorem’s whole yield.

P (Succession: the premise). The tokening of the denial and the tokening of the assertion stand in succession. One instance per realized step; the full run costs an infinite family of instances over pairwise-distinct loci, and the orientation of the chain is part of the premise. P is independent of Th. 1–4: a still structure and a running one share one underlying world and all tokening facts, with P false throughout one and true throughout the other.

Th. 5 (Actuality, from P). Given discrimination and P , the successor’s value is the negation of its predecessor’s: one realized step of the alternation. A chain of instances realizes the run.

Th. 6 (Obstruction). Regiment claims as properties of succession-free worlds. A claim true wherever its denial is discriminated holds at still worlds that discriminate it; given any such still world, the entailment from the claim to running fails. Temporal realization rests on P , and, given such a world, an undeniable substitute for P is unavailable.

Corollary. The alternation exists and is unique up to relabeling of its seed, and nothing static answers to the ground concept (Th. 1–3). A discriminated denial witnesses occurrence (Th. 4). Given P , each such denial realizes a step of the alternation (Th. 5), and P is the exact price of the temporal reading (Th. 6).

Ledger: verified conditionals; five stated identifications (ground as law, structureless as equivariant, ground-hood as exclusion, satisfaction as objectual or processual, claims as world-level); one empirical discharge; the premise schema P .

L1–L3 and Th. 1–4 are named theorems of `lean/Becoming.lean`; Th. 5–6, P , and the independence of P and of the chain hypothesis are in `lean/Obstruction.lean`. Both files are Lean 4, prelude only, with no imports and no axioms, standard ones included.